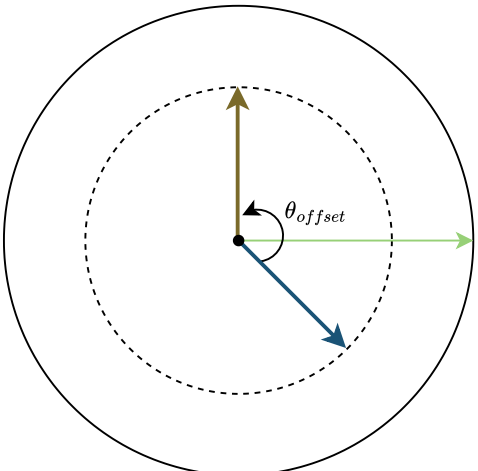
Before Calibration

Encoder reads θ_{enc}



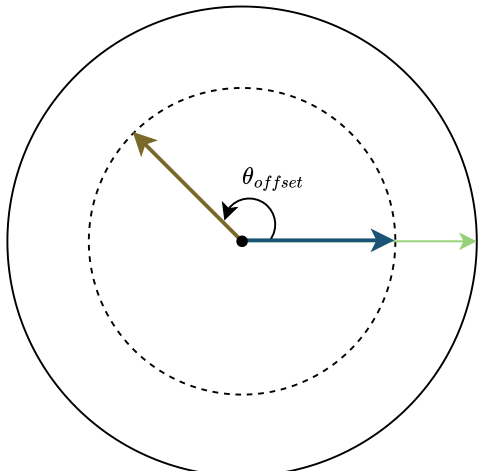
There is an offset angle θ_{offset}

Apply fixed current to lock the rotor

Rotor's $\theta_e = 0$ (d-axis) is unkown

After Calibration

Encoder reads θ_{enc}



The offset angle is now measurable

Rotor's $\theta_e = 0$ (d-axis) is unkown